

Write Units

Constants

$$\begin{array}{ll} h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s} & \hbar = 1.054 \times 10^{-34} \text{ J} \cdot \text{s} \\ h = 4.135 \times 10^{-15} \text{ eV} \cdot \text{s} & \hbar = 6.582 \times 10^{-16} \text{ eV} \cdot \text{s} \\ hc = 1240 \text{ eV} \cdot \text{nm} & \hbar c = 197.3 \text{ eV} \cdot \text{nm} \\ k_B = 1.38 \times 10^{-23} \text{ J/K} & k_B = 8.617 \times 10^{-5} \text{ eV/K} \\ c = 2.998 \times 10^8 \text{ m/s} & e^{\pm} = \pm 1.602 \times 10^{-19} \text{ C} \\ 1 \text{ J} = 6.242 \times 10^{18} \text{ eV} & 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J} \\ m_e = 9.11 \times 10^{-31} \text{ kg} & m_e = 0.511 \text{ MeV}/c^2 \\ \epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2 & \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2 \\ 1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg} & 1 \text{ kg} = 6.022 \times 10^{26} \text{ amu} \end{array}$$

General Waves

$$\begin{aligned} (\nabla^2 \psi) &= \frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} v = \lambda f = \frac{\omega}{k} \\ \psi(x, t) &= \psi_m \cos(kx \mp \omega t); f = \frac{\omega}{2\pi} = \frac{1}{T}; k = \frac{2\pi}{\lambda} \end{aligned}$$

Consine can be exchanged with another function like $\sin(\theta)$ or $e^{i\theta}$.

Superposition of waves also fulfills wave equation.

Pulses tend to follow the Gaussian

$$(e^{-\alpha t}; \int_{-\infty}^{\infty} e^{-\alpha t} dt = \sqrt{\frac{\pi}{\alpha}})$$

Standing waves

Two traveling waves in opposite directions

$$\psi(x, t) = \psi_m \cos(kx + \phi) \cos(\omega t + \phi') = \psi_m (e^{i(kx - \omega t)} + e^{i(kx + \omega t)})$$

Photoelectric Effect Photons hitting electrons beyond a certain energy can cause electrons to break free

$$E = hf = \hbar\omega; K E_{max} = hf - \phi = eV_{stop}; I = \frac{1}{2\mu_0 c} \vec{E}_0^2$$

Bohr Model Proposed quantization of electron angular momentum (1 QN: $rmv = n\hbar$ for $n \in \mathbb{N}$)

Explains emission spectrum of atoms

Derived radius & energy formulas from velocity

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{me^2} n^2; E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2} \frac{1}{n^2} \approx \frac{-13.6\text{eV}}{n^2}$$

Bohr radius a_0 is r_n when $n = 0$

Compton Effect Shoot photon (λ) at electron

Emits photon (λ') at angle θ

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta); \frac{h}{m_e c} = 0.002426 \text{ nm}$$

De Broglie Wavelength: Quantum particles fired at a slit behave like a wave of wavelength $\lambda = \frac{h}{p}$

Schrodinger's 1D Equation: Allows 3 QN

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(x) \right] \psi(x) = E\psi(x)$$

$\vec{F} = -\nabla V$; ψ is probability wave; also used to get averages (function or operator f) or constants

$$P = 1 = \int_{-\infty}^{\infty} \psi^* \psi dx; \langle f \rangle = \int_{-\infty}^{\infty} \psi^* f \psi dx$$

Infinte Well

$$\begin{aligned} V(x) &= \begin{cases} 0 & \text{if } a < x < b \\ \infty & \text{otherwise} \end{cases}; \psi(x) = Ae^{ikx}; k_n^2 = \frac{n^2 \pi^2}{(b-a)^2} = \frac{2mE_n}{\hbar^2} \\ E_n &= \frac{n^2 \pi^2 \hbar^2}{2m(b-a)^2} \approx \frac{n^2 * 54.8 \times 10^{-69} \text{ J}^2 \cdot \text{s}^2}{m(b-a)^2} \approx \frac{n^2 * 4.276 \times 10^{-30} \text{ eV}^2 \cdot \text{s}^2}{m(b-a)^2} \end{aligned}$$

Finite Well Transcendental algebra problem

$$V(x) = \begin{cases} 0 & \text{if } a < x < b \\ V_0 & \text{otherwise} \end{cases}$$

Waves either reflect or transmit at edge

$$T = \frac{4k_1 k_2}{(k_1 + k_2)^2}; R = \frac{(k_1 - k_2)^2}{(k_1 + k_2)^2}; T + R = 1$$

Harmonic Oscillator $V(x)$ follows Taylor series or $V(x) = \frac{1}{2} kx^2$; uses polynomial H_n

$$\psi_n(x) = A_n H_n e^{-ax^2}; a = \frac{\sqrt{km}}{2\hbar}; E_n = \left(n + \frac{1}{2}\right) \hbar \sqrt{\frac{k}{m}}$$

Operators Basically eigenvectors and eigenvalues just with functions and not vectors.

$$\hat{A}\psi = A\psi; \hat{x} = x; \hat{p} = -i\hbar\nabla; \hat{E} = -\frac{\hbar^2}{2m} \nabla^2 + V(x)$$

Heisenberg Uncertainty Principle

$$\Delta x \Delta p \geq \hbar/2; \Delta t \Delta E \geq \hbar/2$$

Fourier Analysis: An extra thing you can use for Laplace transforms

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(k) e^{ikx} dk; g(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(k) e^{-ikx} dk$$

Hydrogen Atom: 3D; Uses spherical coordinates

$$\begin{aligned} x &= r \sin \theta \cos \phi; y = r \sin \theta \sin \phi; z = r \cos \theta \\ r &= \sqrt{x^2 + y^2 + z^2}; \theta = \arccos \frac{z}{r}; \phi = \arctan \frac{y}{x} \end{aligned}$$

Schrodinger's can be turned spherical.

$$\left[\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2}{\partial \theta^2} \right] \psi = E\psi$$

Solution becomes seperable; $\Theta(\theta)$ & $\Phi(\phi)$ in exam

$$\psi(r, \theta, \phi) = R(r) \cdot \Theta(\theta) \cdot \Phi(\phi)$$

Name	Used in	Values	Transition
n	$R(r)$	$\mathbb{N}$	$\mathbb{N}/A$
ℓ	$R(r), \Theta(\theta), \vec{L}$	$\mathbb{N} < n$	$\Delta \ell = \pm 1$
m_ℓ	$\Theta(\theta), \Phi(\phi)$	$\mathbb{N}; -\ell \leq m_\ell \leq \ell$	$\Delta m_\ell \in -1, 0, 1$

Probability functions:

$$P(r) = r^2 |R(r)|^2; P(\theta, \phi) = |\Theta(\theta)\Phi(\phi)|$$

Quantum Angular Momentum ($\vec{L}$)

$$L = \hbar\sqrt{\ell(\ell+1)}; L_z = m_\ell \hbar$$

Zeeman Effect Uses magnetic mield B

$$U = m_L \mu_B B; \mu_B = \frac{e^+ \hbar}{2m_e} \approx 5.788 \times 10^{-9} \text{ eV/G}$$

Many-Electron Atoms

$$E_n = (-13.6 \text{ eV}) \frac{Z_{eff}^2}{n^2}; Z_{eff} = q_{nucleus} - q_{screeners}$$

For transition energies:

$$\Delta E = 13.6 \text{ eV} (Z-1)^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

For electrons in incomplete shells, exists angular (L) & magnetic (M_L) Q.N's:

$$\begin{aligned} |L_i| &= \hbar\sqrt{l_i(l_i+1)}; L_{max} = l_1 + l_2; L_{min} = |l_1 - l_2| \\ L &\in [L_{min}, L_{max}] \cap \mathbb{N}; M_L = \sum m_{l,i}; M_L \in [-L, L] \cap \mathbb{N} \end{aligned}$$

S can be computed same as M_L but with spin quantum number
Hund's rule: $L = M_{L,max}$, $S = M_{S,max}$; maximize total spin
 $\Delta L \in \{0, \pm 1\}$; $\Delta S = 0$
HeNe laser produces photons

$$\lambda = 632.8 \text{ nm}; E = 1.96 \text{ eV}; E_0 = \sqrt{\frac{2I}{c\epsilon_0}}; I = \frac{P}{A}$$

Molecular Structure

$$E(H + H^+) = E(H) + E(H^+) = -13.6 \text{ eV} + 0; E(H_2^+) = -16.3 \text{ eV}; E(H_2) = -31.7 \text{ eV}$$

Covalent bonded atoms share electrons, forming bonding then antibonding
Bond dissociation energy inversely proportional to periodic y-coord and equilibrium separation, proportional to periodic x-coord
Ionic bonds come from 2 ions pulled by Coulombic force
Pauli exclusion prevents 2 atoms from being too close; has potential energy

$$E_{\text{molecule}} = U_R + U_C + \Delta E_{\text{ionization}}; U_C = -\frac{q_1 q_2}{4\pi\epsilon_0 R_{eq}}$$

Can be considered electric dipole of moment p , often off from measured value

$$p = qr = qR_{eq}$$

Vibrational states in molecules:

$$E_N = (N + \frac{1}{2})\hbar\omega; \omega = \sqrt{\frac{k}{m}} = 2\pi f; \Delta N = \pm 1$$

Use harmonic mean for mass

$$m = \left(\sum \frac{1}{m_i}\right)^{-1} \left[= \frac{m_1 \cdot m_2}{m_1 + m_2} \right]$$

Rotational energy also exists, from Q.N. L

$$E_L = \frac{L(L+1)\hbar^2}{2mR_{eq}^2}; B = \frac{\hbar^2}{2mR_{eq}^2}; \Delta L = \pm 1; 2L+1 = \sqrt{\frac{2kT}{B}}$$

$$p(E_{NL}) = (2L+1)e^{-\frac{E_{NL}}{kT}} = (2L+1)e^{-\frac{(N+\frac{1}{2})\hbar f + BL(L+1)}{kT}}$$

Statistical Physics

Indistinguishable particles measure probability of finding one with specific energy E ; Uses microstate weight W_i ; $U(E)$ is energy density at energy E

$$p(E) = \frac{\sum_i N_{E,i} W_i}{N_{\text{total}} \sum_i W_i}$$

$$U(E) = Eg(E)f(E); dN = N(E) dE = Vg(E)f(E) dE$$

Can find density of particles in energy state E

$$g_{\text{mass}}(E) = \frac{4\pi(2s+1)\sqrt{2}(mc^2)^{3/2}}{(hc)^3}; g_\gamma(E) = \frac{E^2}{\pi^2(hc)^3} = \frac{8\pi E^2}{(hc)^3}$$

For dilute gas, go straight to Maxwell-Boltzmann.

$$k = 1.38 \times 10^{-23} \text{ J/K} = 8.617 \times 10^{-5} \text{ eV/K}; K_{avg} = \frac{3}{2}kT$$

$$N(E) = \frac{2N}{\sqrt{\pi}(kT)^{3/2}} E^{\frac{1}{2}} e^{-\frac{E}{kT}}; N(v) = N \sqrt{\frac{2}{\pi}} \left(\frac{m}{kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

Approximate tiny distribution integrals.

$$\int_{E_m}^{E_m + \Delta E} N(E) dE \approx N(E) * \Delta E$$

Doppler Broadening of Spectral Line

$$\Delta f = 2f_0 \sqrt{(2 \ln(2))kT/mc^2}$$

Blackbody radiation energy distribution comes from Bose-Einstein distribution

$$f_{BE}(E) = \frac{1}{A_{BE} e^{E/kT} - 1}; U = \frac{8\pi^5 k_B^4}{15(hc)^3} T^4$$

Fermi-Dirac energy uses Fermi energy E_F

$$f_{FD}(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}; E_F = \frac{\hbar^2}{2m} \left(\frac{3N}{8\pi V}\right)^{\frac{2}{3}}; E_m = \frac{3}{5}E_F$$

Relativity Δt_0 is proper time; L_0 = proper length

$$\Delta t = \frac{\Delta t_0}{\sqrt{1-u^2/c^2}}; L = L_0 \sqrt{1-u^2/c^2}$$

$$v = \frac{v' - u}{1 + v'u/c^2}; f' = f \sqrt{\frac{1-u/c}{1+u/c}}$$

Lorentz Transforms

$$x' = \frac{x - ut}{\sqrt{1-u^2/c^2}}; y' = y; z' = z; t' = \frac{t - (u/c^2)x}{\sqrt{1-u^2/c^2}}$$

$$v'_x = \frac{v_x - u}{1 - v_x u/c^2}; v'_y = \frac{v_y \sqrt{1-u^2/c^2}}{1 - v_x u/c^2}; v'_z = \frac{v_z \sqrt{1-u^2/c^2}}{1 - v_x u/c^2}$$

Relativity alters momentum and energy; total remains constant in any dimension/direction

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1-v^2/c^2}}; E = K + E_0 = \frac{mc^2}{\sqrt{1-v^2/c^2}}; E_0 = mc^2$$

$$E^2 = (pc)^2 + (mc^2)^2; E \approx pc$$

$$v = \frac{p}{\sqrt{m^2 + p^2/c^2}} = c \sqrt{1 - \left(\frac{mc^2}{E}\right)^2}$$